

Formation Dynamics of the Off-Support Extension (MSE vs fadehint)

Task: scripted ToolHang init-to-insertion, 2000 demos, chiunet policy. All statistics are computed per training snapshot of a single run, on fixed query sets. Date: 2026-07-08. Endpoint references: MSE SR 71 (canonical seed), MIP 95, fadehint 96 (twofactor, 100 held-out seeds 21000+).

Column definitions

loss	Training loss (L2 on normalized action chunks) at the snapshot step.
settle% / transit%	The fold statistic. 10-NN phase composition, in the snapshot's embedding space (encoder output, cosine metric), of 108 deployed failure-window states (harvested from converged-MSE failure rollouts, closure window) against a 13k-state phase-labeled dataset bank. Healthy reference: settle 26-36%. Folded: settle ~4%, transit ~85% (the embedding assigns the states to the lift/transit cluster and the policy reads out its labels).
pull-back p50	The recovery-field statistic. Canonical settle states are offset by 10-40 mm in random directions $\mathbf{d}$ (exact kinematics, eef + relative dims). $\text{pull-back} = \mathbf{a}_{\text{pos}} \cdot (-\mathbf{d}_{\text{hat}})$: the component of the commanded action returning toward the data manifold. Positive = restoring; ~ 0 = offset ignored; negative = pushes further out. $+0.099 \sim 5$ mm/step of commanded correction at the OSC gain; $+0.005 \sim \text{none}$. Median over 90 offset states.
align_all / align_settle	On-support metric alignment. Spearman rank agreement between embedding distances and action-label distances over $\sim 30\text{k}$ within-phase pairs of dataset states (all phases pooled / settle phase only). Measures on-support representation geometry.
faith_fail	Off-support metric faithfulness. Mean (over the 108 failure-window queries) Spearman agreement between embedding-distance and physical state-distance rankings from the query to 2000 bank states. High = the embedding orders the world consistently with physical proximity at off-support states; the fold is the collapse of this agreement.
SR	Task success rate, official twofactor harness, 100 held-out seeds (21000+). Auxiliary: cross4 = fraction of rollouts leaving a 4 mm tube around the demonstration manifold; $\text{SR} \text{cross4}$ = success given such an escape (the recovery term); $\text{SR} \text{stay}$ = success given no escape (100% for every snapshot measured).

Table 1 - MSE formation (mse_timeline run; static rows abridged)

step	loss	settle%	transit%	pull-back p50	align_all	align_settle	faith_fail	SR
9k	1.79e-3	16	54	+0.099	0.669	0.605	0.740	42
19k	1.35e-3	24	57	+0.074	0.662	0.675	0.534	45
29k	1.10e-3	13	69	+0.057	0.653	0.723	0.456	—
38k	0.99e-3	5	81	+0.038	0.640	0.750	0.407	36
48k	0.84e-3	4	83	+0.026	0.639	0.764	0.382	—
58k	—	4	82	+0.018	0.637	0.771	0.355	—
68k	0.73e-3	4	85	+0.013	0.630	0.771	0.325	50
78k	—	4	85	+0.015	0.625	0.771	0.310	—
88k	—	4	85	+0.015	0.625	0.766	0.302	—
98k	0.55e-3	4	85	+0.013	0.619	0.768	0.290	—
118k	—	4	85	+0.012	0.613	0.763	0.277	—
148k	1.6e-4	4	85	+0.007	0.597	0.760	0.259	62
177k	—	4	85	+0.007	0.594	0.761	0.270	—
207k	3.5e-5	4	85	+0.008	0.593	0.758	0.272	—
246k	—	4	85	+0.006	0.593	0.761	0.278	—
276k	—	4	85	+0.005	0.593	0.762	0.279	—
300k	4e-6	4	85	+0.005	0.594	0.763	0.279	59

Table 2 - fadehint (orig_fadehint run) + interpretation

Deployment-mode caveat (*)

fadehint trains on $f(s, a + \sigma \cdot \epsilon, 0) \rightarrow a$ with σ log-annealed 0.003 $\rightarrow$ 3.0, and deploys as $f(s, 0, 0)$. The zero-input query is off-distribution until the anneal has taught the network to ignore the action input (weaning completes ~ 240 - 300 k). Starred SR and pull-back values before that point measure curriculum weaning progress, not capability: at 60k the policy wanders (maxd p50 454 mm); at 180k it tracks the tube (maxd p90 5.2 mm) but crawls and times out. Encoder-based columns (settle/transit, align, faith) are valid at all steps. MIP has no such caveat (its 2-step deployment is trained from step one; step-matched MIP table in progress).

Table 2 - fadehint formation (snapshot times mapped to steps via metrics timeline)

step (~)	settle%	transit%	pull-back p50 *	align_all	align_settle	faith_fail	SR *
60k	19	54	(-0.018)*	0.565	0.477	0.829	(0)*
120k	35	54	(+0.095)*	0.521	0.482	0.795	—
180k	35	54	(+0.031)*	0.507	0.510	0.765	(0)* on-tube crawl
239k	35	55	(+0.016)*	0.492	0.563	0.634	—
300k	36	55	+0.018	0.499	0.578	0.634	95 (SR cross4=78)

Findings

1. The benign extension is the early default, not a denoising product

At 9k steps MSE has faith 0.74, pull-back +0.099 (92% of offset states corrected), weak fold. Nothing about the healthy extension is acquired late; it is the property of the smooth early-training function.

2. Plain L2 spends the extension to buy on-support precision

Fold formation and faithfulness collapse coincide (9k-48k, during ordinary fitting; loss only halves in the window). Pull-back continues to bleed through the memorization tail (148k-300k: loss /40, pull-back -30% further). Over the same steps the on-support alignment RISES (0.61 -> 0.76) and loss falls 450x: one quantity (rank-metric quality), two domains, opposite directions.

3. Behavioral cash value: SR = adherence x recovery

SR|stay = 100 at every MSE snapshot; SR|cross4 stays 4-17% at every stage. Training improves adherence (cross4 0.69 -> 0.44, fit-driven) while recovery-from-escape never improves: the depreciated extension is the ceiling. Early extension health is behaviorally worthless without fit (median excursion 42 mm at 9k dwarfs the 10-40 mm annulus where the pull-back operates); hence no interior SR maximum.

4. fadehint keeps the asset and converts it into the recovery term

Endpoint contrast at matched architecture/data/fit-scale: faith 0.63 vs 0.28; pull-back +0.018 vs +0.005; fold settle 36% vs 4%; SR|cross4 = 78% vs 4-17%; SR 95-96 vs 59-71. Additionally fadehint's fold statistic moves opposite to MSE's during training (settle 19% at 60k -> 35% by 120k): sufficient noise pressure actively repairs a partially formed fold - the noise scale is the control knob of the extension's shape.

Caveats

Single training run per objective (seed spreads elsewhere in the record are 10-40 SR points for composite configurations); failure-window queries are converged-MSE-derived (provenance selection); fadehint's early window (0-60k) unmeasured pending the fade_timeline retrain; MIP step-matched table in progress.